

Transformer network for data imputation in electricity demand data

Ashkan Lotfipoor*, Sandhya Patidar, David P. Jenkins

Institute for Infrastructure and Environment, Heriot-Watt University, Edinburgh, EH14 4AS, United Kingdom

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ABSTRACT

Load forecasting necessitates a significant amount of smart meter data. Several elements in this process, including device malfunctions and signal transmission issues, produce missing data gaps. Missing values in the dataset significantly influence the learning ability of machine learning algorithms, and they must be infilled before proceeding with any statistical analysis. This paper investigates the handling of missing values in demand data, and a new approach is developed for improving the performance of demand analytics, such as energy forecasting. The proposed model uses a transformer neural network to impute the missing values at various rates in the demand profile. Our model uses a k-means algorithm to fill in the missing values with proxy values in the dataset. The model is applied to two case-study residential house located in Cornwall and Fintry, United Kingdom. The developed algorithm is assessed for its potential for infilling missing values for three widely understood missing value scenarios: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). The proposed model's imputed outputs are compared to the original dataset to assess model performance. The performance of the framework is compared with a selection of widely used statistical and machine learning models. The proposed transformer model shows significant improvements over the common linear method in all three scenarios (with 30% missing values), with percentage improvements ranging from approximately 49.71% to 57.52% for Cornwall dataset.

1. Introduction

The constant advancement of machine learning and artificial intelligence approaches demonstrates great achievement in a wide range of analytical challenges in many sectors, including the energy industry. Several machine learning-based strategies for precise electric load forecasting have recently been presented. However, load forecasting necessitates a significant amount of smart meter data, which measures the amount of electricity consumed by various appliances. Several elements in this process, including device malfunctions, signal transmission issues, signal processing errors, signal noise, and physical interference can produce missing data gaps. Missing values in the dataset significantly influence the learning ability of machine learning algorithms, and they must be infilled before proceeding with any statistical analysis.

Numerous factors contribute to the prevalence of missing values in electricity demand data. For instance, electricity consumption is measured using meters installed at various locations, such as residential and commercial properties. Due to technical issues or incorrect installation, these sensors can occasionally malfunction or fail to record readings.

Additionally, there are communication issues. Various communication channels can be utilised to transmit data from meters to data collection systems. During this process, network disruptions (especially in households) and signal interference can result in missing values. Most statistical studies require complete data, and researchers have worked to create efficient methods for substituting plausible values for missing information, a process known as data imputation.

In the last two decades, data imputation has received a lot of attention in the literature and the subject is an active area of research [39,40,6,10]. In R [27], one of the primary programming languages for statisticians and data scientists, a wide number of algorithms for missing data imputation have been created. According to a recent survey, there are over 150 implementations available for missing data imputation [21].

In energy modelling, data-driven methods are considered as black-box modelling. Here, the black-box refers to models in which underlying processes leading to outcomes are not directly interpretable [17]. While these methods lack the interpretability, they often excel in capturing complex patterns. Simple data-driven methods such as: hot deck [20], cold deck [5] and mean imputation are the most basic approaches for

* Corresponding author.

E-mail addresses: a1146@hw.ac.uk (A. Lotfipoor), s.patidar@hw.ac.uk (S. Patidar), d.p.jenkins@hw.ac.uk (D.P. Jenkins).

data imputation. Model-based imputation methods employ statistical and machine learning techniques such as: ARIMA, singular value decomposition (SVD), or k-nearest neighbour (KNN) [1,2,26,35]. In recent years, numerous deep learning-based approaches for missing data imputation have been presented, with Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) being the most prevalent. Most recently, [42] proposes a graph convolutional bidirectional recurrent neural network for missing data imputation. In another study, a CNN model is developed for sensor data imputation of dam structural health monitoring systems [18]. [7] proposed a model using various stations of National Climatic Data Center (NCDC) datasets to predict missing data of stations nearest to a geographical station that are having the complete data. Moreover, recent works demonstrate the increased performance of denoising autoencoders for Multiple Imputation compared to existing approaches [16].

The multiplicity of missing data problems implies a wide range of proposed and researched techniques in various domains. The proposed methods in the literature differ in their analysis objectives (estimation of parameters and variance, matrix completeness, prediction), the type of the variables investigated (categorical, mixed, etc.), data assumptions, and missing data procedures [24]. In energy, there are few studies that proposed an imputation framework for energy demand datasets. [19] proposed a bi-directional imputation approach using a Long Short-term Memory (LSTM) network. They test their model with two missing mechanisms: random and continuous. The scope of this experiment is quite narrow, it only considers the demand of a lab building. A robust imputation model needs to be tested on a variety of demand datasets with different patterns. In another study by [14] a hybrid model for data imputation and forecasting of energy demand is proposed. In this method, the demand before and after the missing gap is used in an ensemble model to impute the missing values. [12] developed a scheme utilising a bagging ensemble of multilayer perceptrons (MLPs), called a softmax ensemble network, wherein the ensemble weight of each MLP is determined by a softmax function.

Despite the large range of options, missing data is often not handled appropriately in practice. This may be for a few reasons. First, the imputation model must be tailored to a particular domain. Developing effective machine learning models involves domain expertise, and an imputation model must be created with the specifics of a domain in mind to function. Second, most studies only consider small missing rates. Third, in the existing studies, the only missing value mechanism investigated is missing completely at random (MCAR). However, other mechanisms need to be studied to provide a comprehensive framework for handling missing values.

Most of the imputation approaches are validated through simulation experiments. This validation procedure may have significant drawbacks, such as a lack of generalisability and limited missing data patterns. Instead of a simulated dataset, datasets from real case studies were used in this study to create and test the suggested imputation framework.

To ensure statistical significance, robustness and reliability of the proposed method, the paper covers:

- Data amputation using MCAR, MAR, and MNAR mechanisms;
- Evaluating the performance of the proposed model on two real-world demand datasets in the United Kingdom with different consumption patterns;
- Using a Transformer with self-attention mechanism to process long sequences of data to improve imputation results;
- Using autocorrelation and distribution plots for accessing the performance of methodology in retaining the temporal dynamics of imputed values;

Considering the above contributions, this paper's findings could be used by researchers in energy-related subjects in their future work, and academics familiar with the topic of missing values could benefit from a

detailed data imputation technique and various approaches to missing data generation.

The structure of the paper is as follows: Section 2 gives an overview of key underpinning approaches used for the development of proposed model, clustering, PCA and Transformer model. Section 3 presents the approaches used for data acquisition, data pre-processing, organisation and methodology development of missing data imputation framework. Section 4 explains the experimental set-up of the model, along with baseline methods and error metrics. Then, Section 5 discusses the model performance and key findings. Finally, Section 6 concludes the paper and outlines some potential directions for future research.

2. Basic principles

In this section, the key concepts of clustering, Principal Component Analysis (PCA), and Transformer model, which are used in the proposed method, are discussed briefly.

2.1. Data imputation

There are four steps in a standard approach to data imputation research [30]:

- Step 1: Data collection: For developing an efficient missing value algorithm ideally complete dataset is preferred. This will allow identification of various statistically significant feature/properties of dataset, e.g. mean, variance, percentiles, autocorrelation function (ACF), etc.
- Step 2: Generating synthetic missing values: A percentage of dataset will be set to be missing. Missing values can be generated in univariate or multivariate configurations.
- Step 3: Developing a data infilling algorithm suitable to fit the data type and statistical characteristics noted in Step 1. A range of machine learning and statistical approaches has been used to-date.
- Step 4: Evaluation of imputation algorithms: A comparison of statistical inferences, either in terms of imputation quality or prediction performance, provides an indicator of the missing data method's performance.

Fig. 1 presents these steps in a flowchart. t is the equi-distance and i is the observation in a time series. Black colour in the figure represents the missing values and the green colour represents the imputed values. In this paper, these steps are investigated to develop a robust data imputation method.

2.2. Clustering

k-means clustering [31] is a widely used unsupervised machine learning algorithm that aims to organise a dataset into a number of clusters. The objective of the k-means algorithm is to minimise the within-cluster sum of squared (WCSS):

$$WCSS = \sum_{k=1}^k \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \quad (1)$$

where k represents the number of clusters, C_k represents each cluster, x_i represents a data point belonging to the k^{th} cluster C_k , and μ_k is the mean value of the points assigned to the i^{th} cluster C_i .

k-means is a centroid-based approach, where each cluster being is associated with a centroid. This technique seeks to minimise the sum of distances between each data point and its respective centroid. The algorithm processes the unlabelled dataset, divides the dataset into k clusters and continues the procedure until the optimal clusters are achieved. The k-means algorithm requires a predefined value for k . First, the algorithm assigns a centroid at random to each of the k clusters. The distance between each observation and each of the k centroids is then computed. Observations are then assigned to the nearest centroid. The new position of the centroid is determined by calculating the

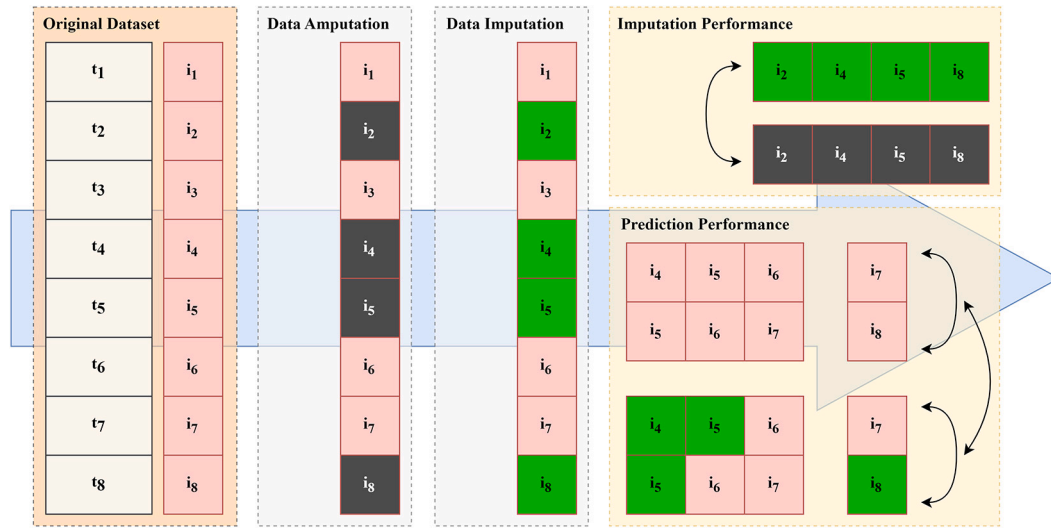


Fig. 1. Experimental setup of an imputation study. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

mean of each cluster's observations. This procedure is repeated until the positions of the centroids do not change. Algorithm 1 presents the inner work of a k-means algorithm.

Algorithm 1 K-means algorithm.

```

1: procedure K-MEANS(data, k, max_iterations)
2:   randomly initialise k cluster centroids from the dataset:  $C = \{c_1, c_2, \dots, c_k\}$ 
3:   for  $i \leftarrow 1$  to max_iterations do
4:     create empty clusters for each centroid:  $C_1 = \{\}, C_2 = \{\}, \dots, C_k = \{\}$ 
5:     for each data point  $p$  do
6:       assign  $p$  to the nearest centroid  $c_j$ :

 $j = \arg \min_j \text{dist}(p, c_j)$ 

7:       add  $p$  to cluster  $C_j$ :  $C_j = C_j \cup \{p\}$ 
8:   for each cluster  $C_j$  do
9:     calculate the new centroid  $c_{\text{new}}$  as the mean of all points assigned to it:

 $c_{\text{new}} = \frac{1}{|C_j|} \sum_{p \in C_j} p$ 

10:    update centroid  $c_j$  with  $c_{\text{new}}$ 
11: return the final cluster assignments and centroids:  $C$ 

```

The primary disadvantage of k-means is that the WCSS converges to a constant value between examples as the number of input dimensions increases. To tackle this issue, it is common to reduce the dimensionality by using principal component analysis (PCA) on the dataset. Another reason to use PCA in clustering is that highly correlated variables might add complexity of modelling structure, and PCA can reduce the number of input variables to mitigate this problem.

There are various approaches to identify the optimum number of clusters, such as elbow plots or average silhouette. Here, the optimum number of clusters in each dataset was identified using the elbow method. The elbow approach works by running k-means clustering on the dataset for a range of k values and then calculating the SSE for each value of k . Elbow method involves plotting the SSE versus the number of clusters. Optimum number of clusters are selected by user for when change in SSE with change in number of clusters is negligible. The details of how a k-mean algorithm is employed in this data imputation framework and what features are used are provided in the next section.

2.3. PCA

Commonly, PCA is applied prior to a clustering process, for transforming high-dimensional correlated features into low-dimensional features while preserving most of the variance in the dataset. Besides being

a technique for dimensionality reduction in clustering, PCA has another attribute termed compressibility in vector clustering [23]. Compressibility shows PCA reduces the distance between data points that belong to the same clusters while reducing inter-cluster distances to a lesser extent.

Given a data matrix containing p correlated variables, PCA transforms them into a set of p uncorrelated components. The first principal components Y_1 are determined by the linear combination of variables $X_1, X_2, \dots, X_p$.

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p \quad (2)$$

The first principal component is computed to accommodate for the largest possible variation. The second principal component is determined as the first, with the condition that it be uncorrelated with (i.e., perpendicular to) the first component and accounts for the next-highest variance. This process is repeated until a total of p main components, equal to the original number of variables, have been computed.

$$Y_p = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p \quad (3)$$

The amount of captured variance decreases with each additional component, though the cumulative variance across all components still sums up to 100%. User can select required number of component depending on their choice of variance tolerance. In this paper, PCA is used for dimensionality reduction in the clustering stage with the PCA function of Python's Sklearn library.

2.4. Attention mechanism

The attention mechanism is a technique used in deep learning to determine the relative significance of input sequence elements when producing predictions. It enables the model to prioritise the most relevant parts of the input when producing a prediction, as opposed to evaluating all elements of the input equally. Attention mechanisms are commonly utilised in natural language processing (NLP) applications such as machine translation and text summarisation, where the order of words in the input sequence is crucial. They are also used in other sequential data tasks such as speech recognition, video analysis and time series forecasting [9,13,38].

There are different types of attention mechanisms, including additive attention, multiplicative attention, and self-attention. The self-attention mechanism is used in the Transformer model, which was introduced in the paper "Attention is All You Need" by [34].

A query and a set of key-value pairs are mapped to output by an attention function, where the query (Q), keys (K), values (V), and output

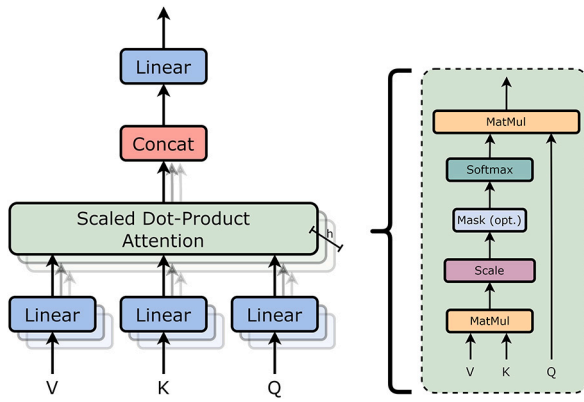


Fig. 2. (Right) Multi-head attention with h attention layers. (Left) Scaled dot-product attention.

are all vectors. The output is generated as a weighted sum of the values, with the weight allocated to each value determined by the query's compatibility function with the relevant key [34]. The attention in a transformer model is known as Scaled Dot-Product Attention. Fig. 2 presents the structure of multi-head attention and scaled dot-product attention. The output matrix of attention is obtained as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

A transformer linearly projects queries, keys, and values h times, enabling the model to simultaneously attend to input from different representation subspaces at different positions. This is known as Multi-head attention, and its calculation is as follows:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (5)$$

where QW_i^Q, KW_i^K, VW_i^V are the weight matrices in queries, keys, values, and output, and h is the number of attention layers.

2.5. Transformer network

The Transformer is a deep learning model that was introduced in the paper “Attention is All You Need” by [34] for machine translation. It is a type of neural network architecture that uses self-attention mechanisms to process sequential data. The Transformer is known for its ability to handle longer sequences and parallelise computations.

The Transformer model consists of two main components: the encoder and the decoder. The encoder takes in the input sequence and produces a set of hidden states, which are then passed to the decoder to generate the output sequence. Both the encoder and the decoder are composed of multiple layers of self-attention and feed-forward neural networks. Fig. 3 shows a transformer architecture. The self-attention mechanism, which was explained beforehand, is the key component of the Transformer and what sets it apart from other neural network architectures.

The feed-forward network is typically composed of two linear layers, with a non-linear activation function in between them. The first linear layer takes in the input and performs a matrix multiplication to map it to a higher dimensional space. The non-linear activation function, such as the linear rectifier unit (ReLU), is then applied to introduce non-linearity in the model. The second linear layer maps the output of the activation function back to the original dimension. The output of the feed-forward network is then concatenated with the input of the self-attention layer and passed through a layer normalisation and again it will be passed to the self-attention layer.

The Transformer also includes positional encoding, which is added to the input sequence to indicate the position of each element in the

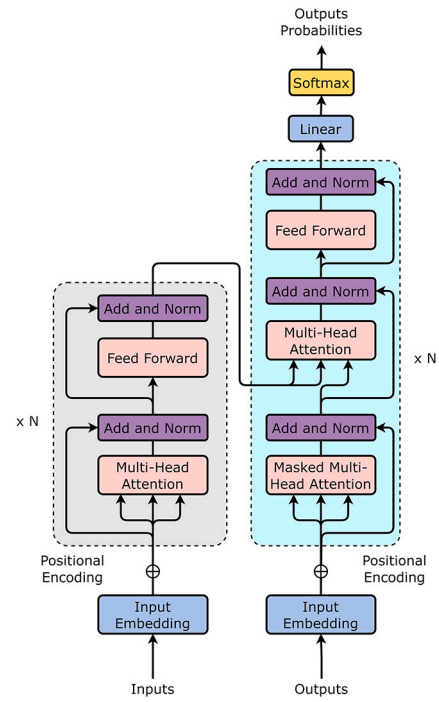


Fig. 3. The transformer architecture.

sequence. This allows the model to take into account the order of the elements in the sequence, which is crucial for tasks such as time series forecasting, where the order of observations is important. Transformers have been used for a variety of tasks recently, namely in image recognition [41], language models [3] and time series forecasting [43].

3. The proposed method

In this paper, we proposed a data imputation system for filling the missing values in energy demand datasets. For developing the proposed model, the consumption from two case study was used. For developing and testing the imputation model, we introduced missing values at different rates with three distinct strategies in the dataset. Then, the time series is transformed for developing a regression model to fill the dataset with forecasted values. Samples with the known y were used for training the model and the samples with missed y were used for testing. The model's procedure is illustrated by the flowchart in Fig. 4. However, in order to reduce error further, the model employs a component that uses proxy values to fill in the missing values in the input space. Inspired by the work of [7], we use values from households with similar demand patterns as proxy values in this case. To accomplish this, all households in each case study are first clustered. The average consumption of each cluster is then computed. Finally, values from the average consumption of associated cluster are used to fill missing values in the input space in the transformed dataset used for data imputation. Fig. 5 depicts a schematic representation of this process.

3.1. Data acquisition and pre-processing

To thoroughly validate the imputation framework suggested in this paper, comprehensive experiments on two distinct energy demand datasets were conducted. These two case studies are referred to as the Cornwall dataset, and the Fintry dataset, respectively. The Fintry dataset is the energy demand data that is collected in Fintry, Scotland. Fintry is a small community in the Scottish county of Stirlingshire. Fintry contains approximately 350 houses. During the period of May 2016 to April 2018, the SMART Fintry project, which was funded by the Scottish Government Local Energy Challenge Fund (LECF), collected

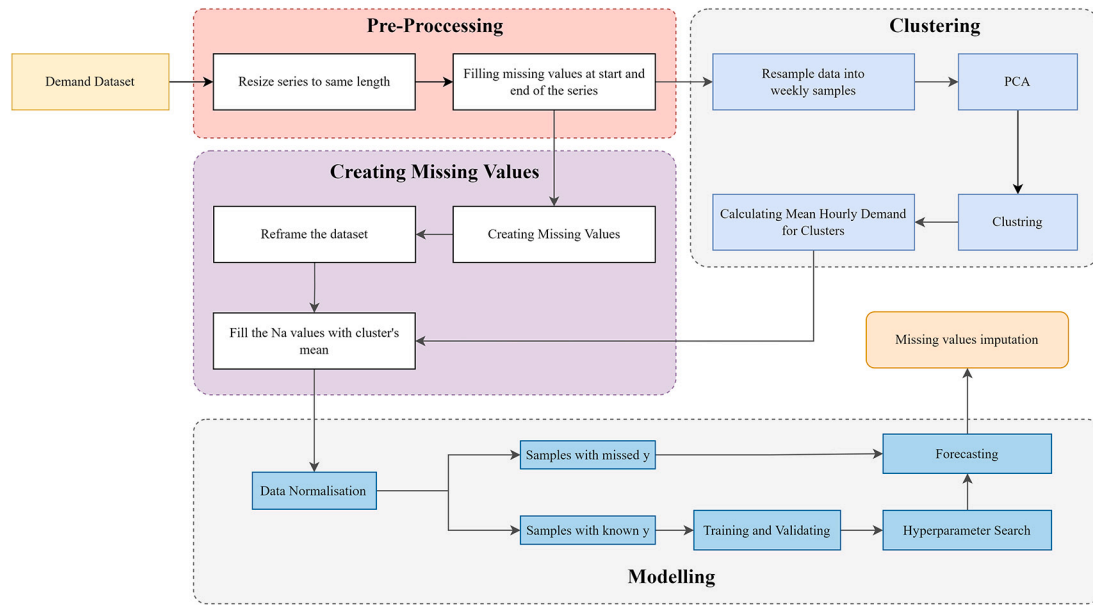


Fig. 4. Proposed imputation framework.

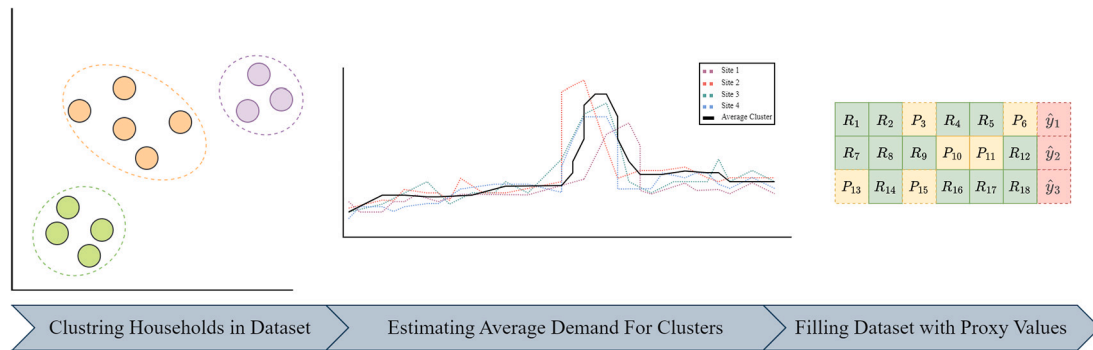


Fig. 5. Filling input space with proxy values.

Table 1
Characteristic of observed datasets.

Project Name	Fintry Dataset	Cornwall Dataset
Number of Sites	48	87
Collection Period	January 2017 to September 2017	April 2014 to September 2015
Number of Observations	15492	158689
Data Resolution	30 Minute	5 Minute

electricity demand data from 115 homes in the community [32]. 48 sites were chosen from the 115 households collected for this study. After processing, each dataset has 15480 observations at a 30-minute time interval.

The next dataset is collected as part of The Cornwall Local Energy Market (LEM), a £17 million project that aimed to design, construct, and test an auction-based flexibility market for distribution network-connected decentralised energy resources (DER) [25]. Data was gathered from 100 residential residences spread over Cornwall. This project collected three types of datasets: measurement datasets, forecast datasets, and dimension datasets. In this research, measurement datasets are used to test the proposed model. The Cornwall dataset was resampled to 5-minutely data for faster modelling. Details of each dataset are presented in Table 1.

To ensure a thorough validation of the model's performance, the two case studies were carefully selected. These case studies involved two distinct locations in the United Kingdom with contrasting characteristics. One dataset represented a small community, while the other

consisted of households spread across Cornwall county. These sites were chosen to encompass different usage patterns and observation resolutions, thereby providing a robust test for the model's generalisation ability. Another crucial factor behind the selection of these datasets was their high data quality. Both datasets exhibited an exceptional characteristic of having minimal missing values in the load data of most households. This enabled the deliberate introduction of missing values for testing purposes, evaluating the accuracy of the model and comparing the distribution of imputed data with the original measurements.

However, a number of sites had to be excluded from this study due to poor data quality, primarily because of excessive missing values. It's important to highlight that for evaluating the data imputation methodology, only one site from each dataset was utilized. Nevertheless, during the clustering stage, all sites were included. The two households selected for imputation had the advantage of having zero missing values. Another significant factor in choosing these two households for the study was their cluster size, as they belonged to the most prominent

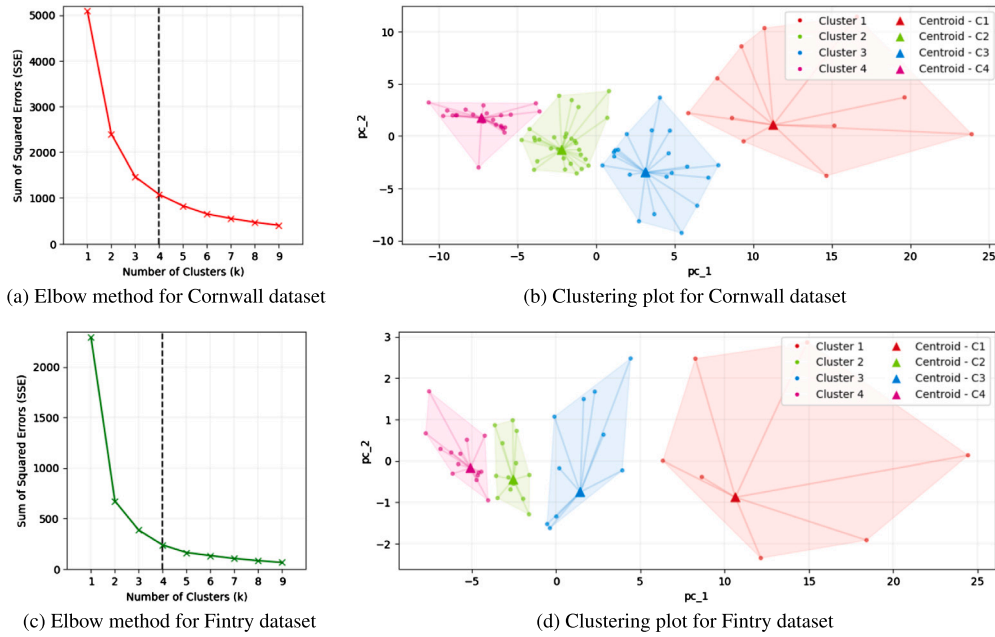


Fig. 6. Elbow method plots (left panel) and k-means clustering plot (right panel).

cluster in each case study. The threshold for the ratio of missing values allowed in datasets used for clustering was set at 5%.

The datasets in the two cited case studies are collected for several households over extended time periods. The initial stage in pre-processing is to resize the demand data for each dwelling into comparable time frames. Since the value range of the raw demand data varies, it is necessary to normalise the value range. This is essential for training a deep learning model for forecasting because with normalisation features contribute proportionally to the model. A second reason for using feature scaling is that gradient descent converges substantially more quickly with feature scaling than without [36]. Finally, it has been demonstrated that standardisation before the clustering process delivers a higher quality, more efficient, and more accurate clustering output [22]. The datasets in this study were normalised using the Z-score normalisation technique. The formula for calculation is given in Equation (6):

$$Z = \frac{(x - \mu)}{\sigma} \quad (6)$$

Where x is the observation and μ and σ represent the mean and standard deviation values of the original demand data, respectively. Both the training and test data sets will be subjected to this normalisation process.

3.2. Clustering stage

This stage's primary purpose is to cluster sites with similar demand patterns and estimate the average consumption of each cluster in order to replace the missing values in the input features of the imputation model with proxy values. The result of this stage is also used as the input in the MAR amputation logistic model, which is presented in the next section.

As previously stated, for clustering with the k-means algorithm, the number of input dimensions must be small. In order to accomplish this, each dataset was resampled to a weekly resolution. Then, using PCA the number of features was reduced further. There are numerous methods for determining the number of principal components to retain in PCA, including scree plot, proportion of total variance explained, average eigenvalue rule, and Log-eigenvalue diagram [4]. The scree plot approach includes plotting the eigenvalues of each component against its corresponding component number in descending order. The "elbow"

point on the plot is used as a cutoff point for determining the number of components. Here, using the scree plot the number of principal components was selected as seven. Finally, these seven principal components were used as grouping features/characteristics in the clustering algorithm.

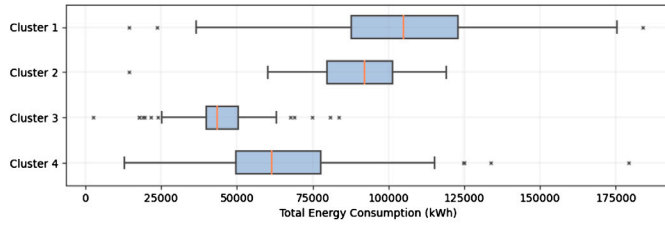
Because of the algorithm's dependency on initialisation, it is necessary to create multiple replicates before selecting one based on the WCSS. The mean silhouette index is the ratio of compactness and separation of a cluster. This metric can evaluate the clustering results. For each iteration, 10 instances of k-means were generated to identify the optimal instance. The mean silhouette index was utilised to find the optimal instance. The clustering process was implemented in Python 3.6 using the k-means function from the Sklearn module.

For both case studies, the number of clusters is selected as four. The elbow plot does not show any significant decrease after the 4 clusters. Corresponding plots for the elbow method for each dataset are presented in Fig. 6. As demonstrated in the plot, a clear clustering is accomplished with no cluster overlap. Fig. 7 presents the boxplot of total energy consumption for each cluster in the two case studies. This plot demonstrates key statistics about the clusters in each dataset, the minimum, maximum, median, first and third quartiles.

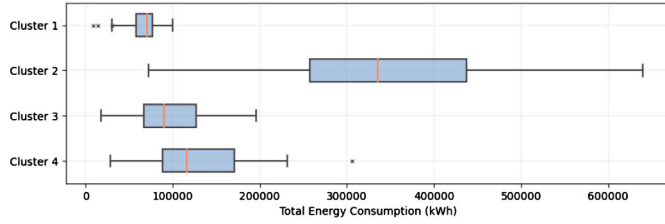
From the plots, it is evident that there is substantial variation between the two case study clusters. Each cluster in the Fintry case study reveals a distinct consumption pattern in the dwellings. Cluster two has the highest dispersion in its demand readings. Only clusters one and four contain outliers in their respective datasets. In contrast, in the Cornwall dataset, all four groups contain outliers. The cluster one case study in Cornwall displays a normal distribution. The distribution is symmetric when the median is in the centre of the box and the whiskers are roughly the same on both sides of the box. The other three clusters are skewed.

3.3. Generating missing values

The statistical relationship between observations (variables) and the probability of missing data is captured by missing data mechanisms. [28] categorises the causes of missing data into three missing data mechanisms: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). For another classification of these mechanisms we can refer to [8].



(a) Boxplot of Cornwall total consumption over 4 cluster



(b) Boxplot of Fintry total consumption over 4 cluster

Fig. 7. Box plot of total energy consumption across four clusters in each case study.

Under MCAR, the probability of missing data is unrelated to the observed or unobserved data. In simpler terms, the occurrence of missingness is entirely independent of the data itself, rendering it random in nature. A broader category of MCAR is MAR, where the probability of an observation being missing depends solely on the observed data. In this case, the missingness can be predicted or explained by the available information in the dataset. On the other hand, MNAR indicates that the probability of missing data varies for reasons that are unknown or not captured by the observed data. This mechanism suggests a non-random pattern of missingness, where the missingness itself may be related to the unobserved values or factors. MNAR poses a more complex challenge, as the missingness mechanism cannot be directly modeled based on the observed data.

The complete observations are denoted by $\mathbf{X} \in \mathcal{X}_1 \times \dots \times \mathcal{X}_p$. Here, it is assumed that $\mathbf{X}$ is a tuple of p columns $X_j \in \mathcal{X}_j$, $j \in \{1, \dots, p\}$, where $\dim(\mathcal{X}_j) = n$ for all j . The missingness pattern $\mathbf{R}$ of the observations $\mathbf{X}$ is an indicator matrix $\mathbf{R} \in \{0, 1\}^{n \times p}$ such that $R_{ij} = 1$ if X_{ij} is observed and $R_{ij} = 0$ otherwise. According to this matrix, the observations $\mathbf{X}$ can be divided into observed and missing: $\mathbf{X} = (\mathbf{X}^{obs}, \mathbf{X}^{mis})$. Let's assume that both $\mathbf{X}$ and $\mathbf{R}$ are random variables with probability of $\mathbb{P}_X$ and $\mathbb{P}_R$. And missingness distribution $\mathbb{P}_R$ is parameterised by ϕ [21]. Now, the probability of a value being missing with the MCAR mechanism is written as follows:

$$\mathbb{P}_R(\mathbf{R} | \mathbf{X}^{obs}, \mathbf{X}^{mis}; \phi) = \mathbb{P}_R(\mathbf{R}) \quad \forall \phi. \quad (7)$$

And the probability that values are missing with the MAR mechanism is written as:

$$\mathbf{X}^{mis} . \mathbb{P}_R(\mathbf{R} | \mathbf{X}^{obs}, \mathbf{X}^{mis}; \phi) = \mathbb{P}_R(\mathbf{R} | \mathbf{X}^{obs}; \phi) \quad \forall \phi, \forall \mathbf{X}^{mis}. \quad (8)$$

We employed all three mechanisms that underlie missing values in a dataset to evaluate the model under real-world settings. We created three functions for generating missing values in complete datasets, which we refer to in this paper as amputation. These functions are widely based on the code published in a paper by [24]. Finally, these functions were wrapped inside the amputation algorithm to generate missing values in datasets. The pseudocode in Algorithm 2 presents the amputation algorithm.

For MCAR, using a Bernoulli distribution of a p_{miss} parameter, which is the proportion of missing values for the target variable, missing values are induced in the dataset. For the MAR mechanism, missing values are generated using a logistic model. According to the MAR definition, the probability of a missing variable is contingent on the observed data. Using the cluster average demand as the observed data, we generate

Input Features								Target Value
lag_48	lag_47	...	lag_4	lag_3	lag_2	lag_1		target
0.29	0.72	...	0.81	0.19	0.292	0.46		0.42
0.53	0.33	...	0.12	0.6	0.79	0.45		NA
...	...	...	...	...	...	...		...
0.292	0.12	...	0.53	0.62	0.12	0.61		0.63
...	...	...	...	...	...	...		...

Fig. 8. Data preparation.

missing demand values. Prior to the amputation procedure, households are clustered and the average energy usage for each cluster is obtained. Then, missing values are induced in the building's energy consumption by applying a logistic model to the cluster's average demand as the observed data.

A logistic model using missing values as variables is finally built for the MNAR. This method is identical to the MAR process mentioned previously, except that the model's inputs may have missing values. Here, the missing probabilities of the energy demand are selected using a logistic model with two input variables, the energy demand (missing values are permitted) and the average demand of the cluster. Variables are separated into a set of inputs for a logistic model and a set for which the logistic model determines the missing probabilities. Inputs are then masked MCAR (hence, missing values from the second set will depend on masked values). Here, the weights are chosen at random, and the intercept is chosen to achieve the desired percentage of missing data.

Algorithm 2 Generating missing values.

```

1:  $X$  = demand dataset
2:  $p_{miss}$  = ratio of missing values
3:  $mechanism$  = "MCAR", "MAR", "MNAR"
4:  $p_{obs}$  = proportion of variables used for the logistic masking
5:
6: function AMPUTATION( $X, p_{miss}, mechanism, p_{obs}$ )
7:   if  $mechanism$  is MAR then
8:     mask with logistic masking model ( $X, p_{miss}, p_{obs}$ )
9:   else if  $mechanism$  is MNAR then
10:    mask with logistic masking model ( $X, p_{miss}, p_{obs}$ )
11:   else  $mechanism$  is MCAR
12:     mask using a Bernoulli distribution of a  $p_{miss}$ 
13:   return  $X_{incomplete} \leftarrow mask.bool(true) = NA$ 

```

3.4. Data preparation

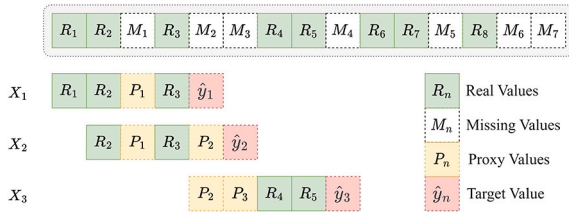
After the clustering and data amputation, each dataset needs to be prepared for the modelling stage. Fig. 8 depicts the output of this stage. For the purpose of this study, one site from each case study is selected and is used for developing the imputation model. After generating the missing values in the demand data, each dataset needs to be transformed into a shape suitable for the imputation with a machine learning algorithm. Using a rolling window method, each dataset is transformed into the sample set for the model. Each time series signal is transformed to a matrix of $n \times m$ shape, where n is the number of samples and m is the number of lag observations used as features. The last column of the matrix is considered as the target value, which needs to be predicted via the model. Rows of the data with known y are considered as the training sample for the model and samples with unknown y are considered as the test sample. Finally, the missing values in the input space will be filled with proxy values (mean demand from each cluster). Fig. 9 illustrates the data preparation procedure in this study.

After creating a matrix similar to the one presented in Fig. 8, each dataset is ready to be imputed using the developed imputation model. The details of the model are presented in the next section.

Table 2

Comparison of RMSE for case study I: Cornwall LEM, for different imputation approaches.

Mechanism	mr	Statistical Methods		Imputation Packages		Machine Learning				Transformer
		Mean	Linear	SoftImpute	Mice	SVM	RF	MLP	CNN-LSTM	
MCAR	0.1	41.05	38.25	37.66	27.22	31.36	25.68	28.66	24.19	18.63
	0.2	41.08	39.94	37.90	30.70	35.27	28.38	30.52	29.04	20.44
	0.3	50.68	48.40	42.57	31.37	37.11	30.55	40.32	29.55	20.62
MAR	0.1	45.93	41.46	37.35	33.54	36.35	29.03	32.91	26.38	18.95
	0.2	46.93	42.80	41.32	36.30	39.90	33.20	33.79	28.32	20.95
	0.3	51.48	48.61	45.24	40.54	42.68	35.69	40.22	31.15	22.91
MNAR	0.1	46.39	45.69	38.04	37.73	41.40	31.23	32.04	28.94	19.61
	0.2	51.61	46.01	45.50	40.37	41.51	37.79	33.43	30.41	22.29
	0.3	56.19	49.09	48.24	41.09	43.74	36.12	36.99	32.22	24.39

**Fig. 9.** Data preparation.

4. Experimental design

All validations are conducted on a workstation with an Intel Core i7 processor and Nvidia RTX 3080 Ti GPU with 64 GB of Ram. The suggested model is written and developed using Keras's functional API with TensorFlow 2.1.0 backend in Python 3.6.

To tune the transformer model using Grid Search with the validation set's performance as the deciding factor, the multi-head attention head number was set between 8 and 16, the encoder and decoder layers were set between 2, 3, 4, 5, and 6, and the output dimension was set between 16, 32, 64, 128, 256, and 512. We set the number of encoder layers to eight, the number of decoder layers to six, the dimension of the model to sixty-four, the number of heads to eight, the number of hidden states in the feed-forward layer to 512, and the dimension of attention q, k, and v to eight. Our model was optimised using the Adam optimizer [15], and the learning rate is 1e-5. For optimal generalisation performance, 150 epochs with early stopping were utilised. A batch of size 128 was used for training. Besides, a dropout of 0.2 was applied to our model.

We compared our Transformer model to several different imputation models. There are three types of models compared. Statistical methods: Mean, Linear; Imputation Packages common between practitioners: Multiple Imputation by Chained Equations (MICE), SoftImpute; Machine Learning: K-nearest neighbour (KNN), Random Forest (RF), MLP, CNN-LSTM. When using the mean technique, missing values are substituted with the mean of non-missing observations. For the Linear method, a linear interpolation algorithm was used. MICE is one of the most often used data imputation methods, and it is frequently used as a baseline for evaluating the effectiveness of new methodologies. Here for implementing the MICE method, the implementation of this technique in MiceForest package [37] was used. This package imputes missing values using LightGBM in an iterative method known as MICE.

SoftImpute [11] is a software package in R for missing values imputation. This technique performs missing values imputation using iterative soft-thresholded SVD's. This technique is based on a low-rank assumption for the data and is justified by the fact that many large matrices are approximated by a low-rank structure [33]. We use an implementation of this algorithm by the FancyImpute package in Python [29]. Additionally, we compare our method to four ML-based methods.

For all ML methods, the input length for the model is 48 steps and the step of predict length is one. For CNN-LSTM, we set a dense connected network and a stack of CNN and LSTM layers. The data was fed into the CNN-LSTM layer for learning historical sequential information, and the final output from it was fed into dense connected layer to fit the number of steps for target prediction. For MLP, we used dense connected network as the basic component. For SVM and RF using random search the optimum hyperparameter were selected and final outputs were obtained using cross validation.

In this paper, for evaluation of the model accuracy root mean squared error (RMSE) are utilised:

$$RMSE = \sqrt{\left(\frac{1}{n}\right) \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

Where y_i is the real consumption, and $\hat{y}_i$ is imputed consumption.

5. Results and discussion

The results of the imputation model evaluation for each case study are presented for two case studies for three missingness mechanisms, MCAR, MAR, and MNAR.

5.1. Case study I: the Cornwall LEM

Following the development of the imputation model, the model was trained and evaluated using the Cornwall dataset. A household was chosen from 87 sites for this purpose. As described previously, the test set for this study was created by introducing artificially-missed observations. In order to train the model, these observations in the input dataset were later replaced with proxy values. The imputation performance of the proposed method is examined under three distinct mechanisms for missing values and three different missing rates (mr), 10%, 20%, 30%. Table 2 shows the imputation error for two statistical imputation methods, two imputation packages in python, four machine learning methods and the proposed transformer approach. These results are also presented in Table 2.

Table 2 demonstrates that, on average, machine learning algorithms outperform statistical approaches. This is attributed to their capacity to incrementally learn high-level features from data. The dataset using MCAR mechanism has the lowest imputation error, whereas MNAR mechanism has the largest imputation error. This is supported by the fact that the MCAR mechanism for missing data is the hardest to estimate, as the missingness is not random. As expected, the performance of all imputation methods decreases as the proportion of missing values increases. This becomes more apparent between %20 and %30. Yet, the proposed technique has the least performance gap between the three missing rates. This demonstrates that the model is capable of producing excellent outcomes, regardless of the missing rate. The RMSE error rate is almost constant between three different missing rates. Another point

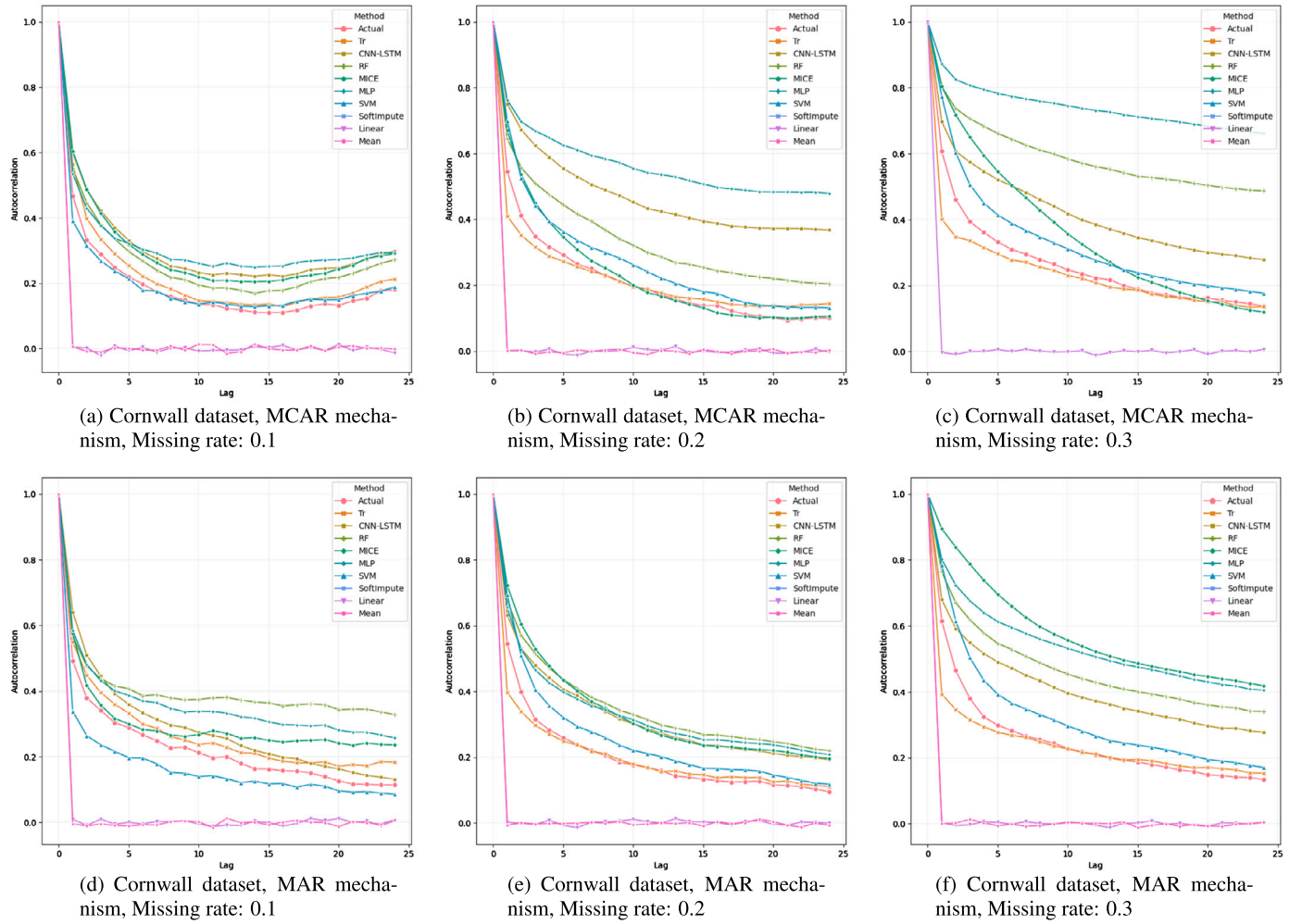


Fig. 10. Autocorrelation plot for two missingness mechanism and three different missing rates, Cornwall dataset.

to note is that imputation libraries in Python such as MICE and SoftImpute demonstrate good performance across the board. This is more evident in the performance of MICE, which use the Python LightGBM algorithm. Fig. 12 presents the RMSE value for all missing rates and missingness mechanisms.

In addition, the transformer outperforms all other approaches investigated for this project. The advantages of the proposed strategy can be justified as follows: The attention mechanism in the transformer model is an unique method for predicting outcomes from long sequences of input data. More precisely, the timestep with the most significant impact on target value (missing value) receives the most attention. In addition, the clustering stage of the proposed method replaces missing values in the input data with proxy values, which, in comparison to the method employed by other approaches (Linear Interpolation), greatly improves the model's ability to accurately impute the data. Another advantage of the transformer that is not evident from Table 2 is that the training and testing speed of a transformer network is considerably faster than that of a CNN-LSTM model. This is because the transformer is capable of parallel computation using GPUs, which substantially accelerates development.

The primary objective of an imputation model is to retain the data's structure, not to accurately predict the true value of the missing observations. This is because, in real-world situations, the true value of missing observations is not available to compare the results. Consequently, the accuracy of the model is not the goal; rather, the correlation between the observations after imputation must be maintained. However, this is only true for MCAR and MAR missing values, while the preserva-

tion of data structure for MNAR values indicates inferior imputation. In order to determine if the imputed values have the same correlation, which is the primary purpose of data imputation for time series datasets, the autocorrelation plot of the imputed values by the methods discussed is plotted. The Fig. 10 presents two sets of plots. The upper panels present the autocorrelation plot of MCAR for three missing rates. The lower panel presents the same plots for MAR.

This figure indicates that the output of the proposed transformer model corresponds most closely to the autocorrelation plot of actual values. Additionally, the same pattern of behaviour is observed amongst the various missingness mechanisms, with the MCAR mechanism producing more accurate results than the MAR method. The worst outcomes are generated by the mean and linear imputer, where there are no correlations after lag one. Moreover, when the rate of missing values increases, the ability of models to replicate the pattern of actual values becomes increasingly challenging.

To study further the preservation of data structure in results, the distribution plot of the imputed dataset versus the actual values is plotted and studied. Here, the most perfect imputation has a distribution function similar to that of the actual values. Fig. 11 demonstrates that the distribution of the transformer model is the closest to the actual data. The same behaviour is seen for both mechanism. In spite of the fact that several machine learning algorithms have acceptable error ranges in this imputation task, their capacity to capture the relation between observations is not equal to that of the transformer. This is evident by the result of autocorrelation and distribution plots.

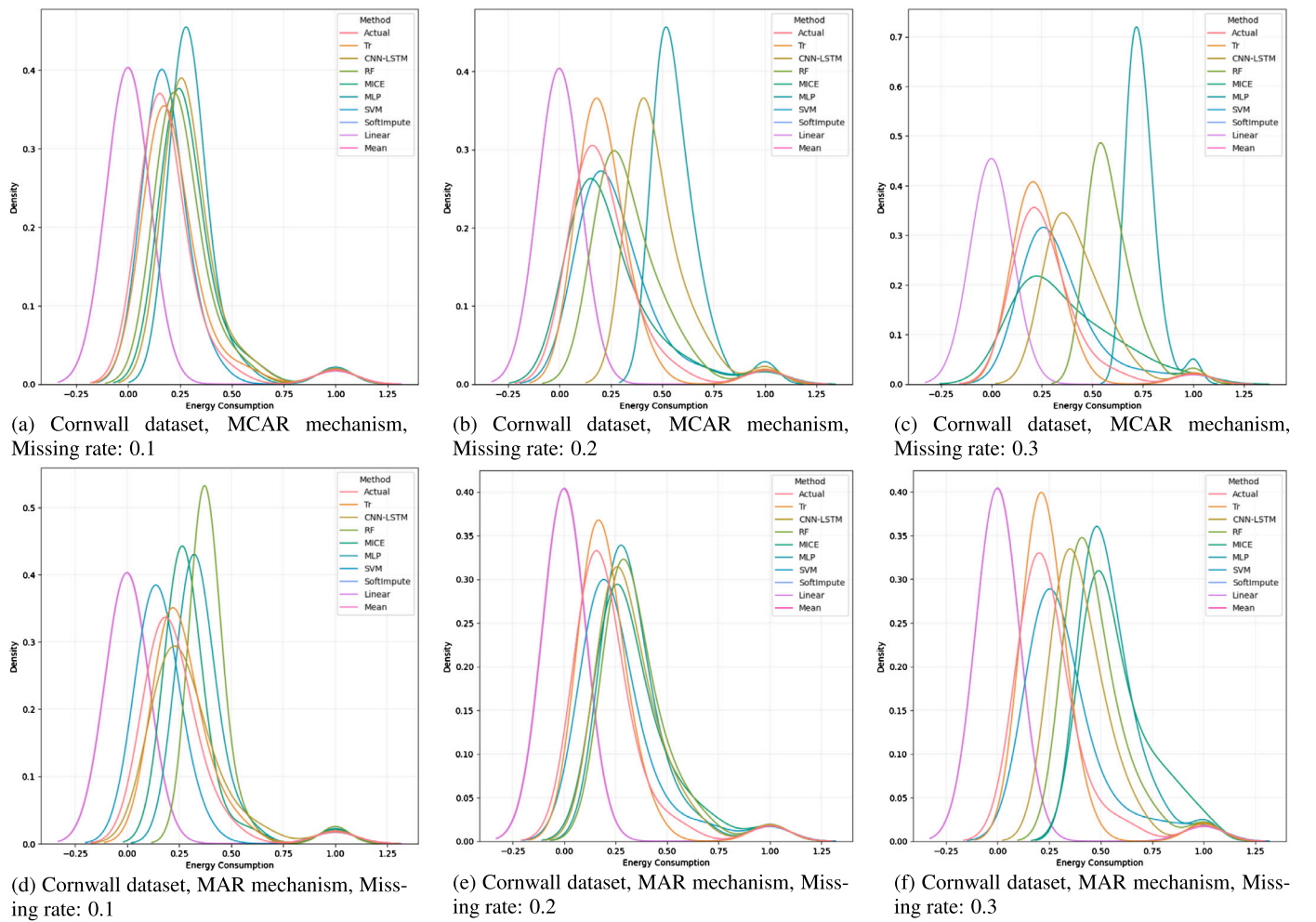


Fig. 11. Distribution plot for two missingness mechanism and three different missing rates, Cornwall dataset.

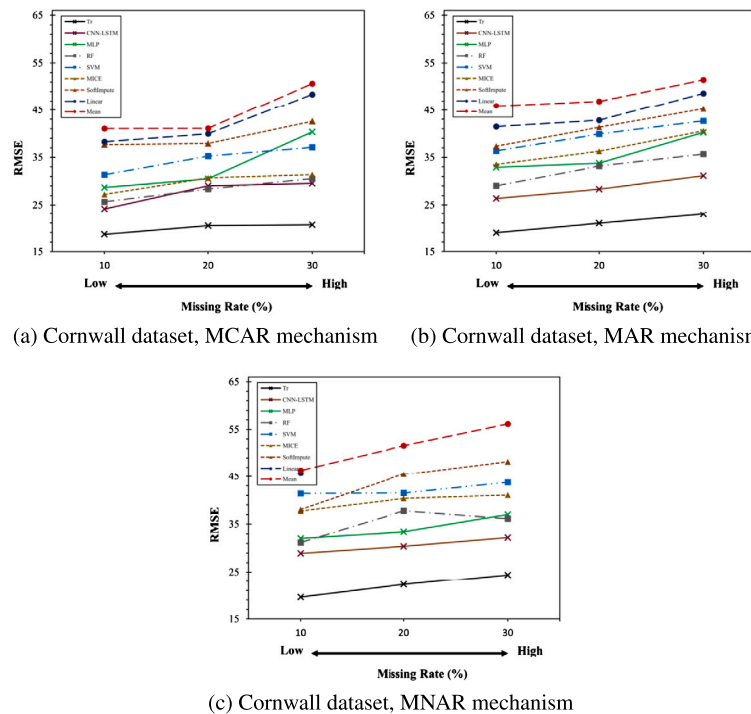


Fig. 12. RMSE values for two missingness mechanism and three different missing rates, Cornwall dataset.

Table 3

Comparison of RMSE for case study II: Fintry, for different imputation approaches.

Mechanism	mr	Statistical Methods		Imputation Packages		Machine Learning				Transformer
		Mean	Linear	SoftImpute	Mice	SVM	RF	ANN	CNN-LSTM	
MCAR	0.1	1143.46	523.75	522.12	374.40	365.22	439.13	333.35	327.64	276.28
	0.2	1166.15	559.06	549.32	399.88	375.55	481.81	354.75	356.16	288.36
	0.3	1143.67	587.65	570.78	415.06	384.87	508.18	369.95	376.23	314.83
MAR	0.1	1159.85	692.84	652.05	558.32	427.03	498.02	406.40	409.94	336.79
	0.2	1235.27	696.38	654.15	562.51	438.76	488.55	412.03	423.94	345.96
	0.3	1339.27	697.50	655.76	487.49	431.00	572.44	429.07	430.90	357.31
MNAR	0.1	1008.35	951.80	987.69	735.75	586.04	708.63	546.80	543.44	432.54
	0.2	1086.92	991.72	918.36	755.26	572.96	678.91	545.66	546.35	475.09
	0.3	1105.73	1067.15	870.00	710.96	556.38	686.37	539.50	539.91	482.56

5.2. Case study II: the Fintry village

As explained before, to fully test the proposed methodology, two datasets with completely different demand pattern were selected to train and test this approach. For the second case study a household in Fintry village was selected. Similar to the last case study, after pre-processing the data and generating missing values in dataset, the missing values in impute data were substituted with proxy values. Next, using the nine imputation techniques mentioned beforehand, the data imputation process was performed. Table 3 presents the result of this study.

Similar to the first case study, the transformer model outperforms existing data imputation methodologies for Fintry demand profile. The MLP and CNN-LSTM models demonstrated promising results, however in comparison to the transformer model, the computation cost of running a CNN-LSTM model is too high for its practical application. In addition, in no circumstance does the MLP model match the accuracy of the transformer model. Of the examined methodologies, statistical techniques, mean imputation and linear interpolation, perform the worst. Machine learning methodologies develop a non-linear representation of the problem, which fits better to real-world datasets.

Considering the capability of the model to capture the structure of the dataset, Fig. 13 depicts the autocorrelation plot for the Fintry dataset. Similar to the previous case study, the transformer model captures the correlation of the original dataset superior to all other imputation techniques. CNN-LSTM model comes in second. The statistical methods fail to capture the demand profile pattern in this instance as well.

6. Conclusion

In conclusion, a novel transformer network is developed in this paper for data imputation in energy demand profiles. This study improves previous work, namely in terms of precision and missing value procedures. Here, a transformer network and a pre-processing pipeline for generating missing values are designed in order to fill missing values for three distinct missingness mechanisms (MCAR, MAR, MNAR). This framework can enhance the accuracy of energy demand imputation tasks. The model is trained and evaluated using data from Fintry and Cornwall, two communities in the United Kingdom. In terms of RMSE, the proposed technique outperforms all alternatives with three different missing rates. In Cornwall case study, for MCAR, MAR, and MNAR scenarios with 30% missing values, which is the highest tested missing rate in this study, the proposed technique shows 57.52%, 49.71%, and 50.21% respectively, compare to the linear method. In Fintry case study, these improvements are 46.45% for MCAR, 48.71% for MAR, and 54.87% for MNAR.

In addition, we established that the autocorrelation between the imputed values and the actual values was quite comparable. Furthermore, it was demonstrated that the distribution of the imputed values

by the transformer corresponds to the actual values more closely than any other approach.

The implications of this research are noteworthy. The improved data imputation method offers several benefits to end-users, such as those in Fintry or Cornwall, energy providers, and policy makers. With a more complete dataset, they can derive more accurate insights and make informed decisions related to energy demand analysis and planning. The ability to fill missing values effectively and obtain reliable imputed data empowers stakeholders to extract valuable information and identify trends, patterns, and anomalies in energy demand profiles with greater confidence. This can be particularly valuable for researchers involved in energy planning, load forecasting, infrastructure optimisation, and resource allocation.

The limited transferability of this method to other cases is one of its limitations, as the proposed methodology requires community-level data that may not be available. However, the computational and data requirements associated with the developed model are justified by the added value it brings to the data. It is also crucial to acknowledge that the time required for dataset development and imputation is a significant aspect of this study. While yielding substantial benefits, the efficiency of the proposed approach should be considered in practical applications. The improved precision in imputing missing values ensures a more accurate representation of the energy demand profiles, leading to more reliable analyses and decision-making. Furthermore, the availability of more comprehensive data facilitates the evaluation of energy-saving initiatives, allowing stakeholders to assess their effectiveness and make data-driven adjustments. Statistical methods are an alternative to the use of clustering for estimating proxy values in this method. Statistical methods can be used to fill in the missing gaps in the input space. However, this will result in less accurate imputation.

Future work will involve two distinct directions. Initially, there are other methods for amputing the data by different algorithms. For Example, for MNAR, there are self-masked and quantile censorship. Generating simulated missing values is a crucial aspect of data imputation research, thus these algorithms must be examined to determine if the model has a similar performance on them as well. Like with all statistical research, there is the experiment's scope. In future research, expanding the scope of the experiment to include a greater number of datasets from various types of houses and communities would further validate the model's performance and applicability. This broader exploration would enable a better understanding of how the proposed approach can be effectively applied in different contexts and scenarios.

CRedit authorship contribution statement

Ashkan Lotfipoor: Conceptualization, Data curation, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Sandhya Patidar:** Conceptualization, Methodology, Writing – review & editing. **David P. Jenkins:** Conceptualization, Writing – review & editing.

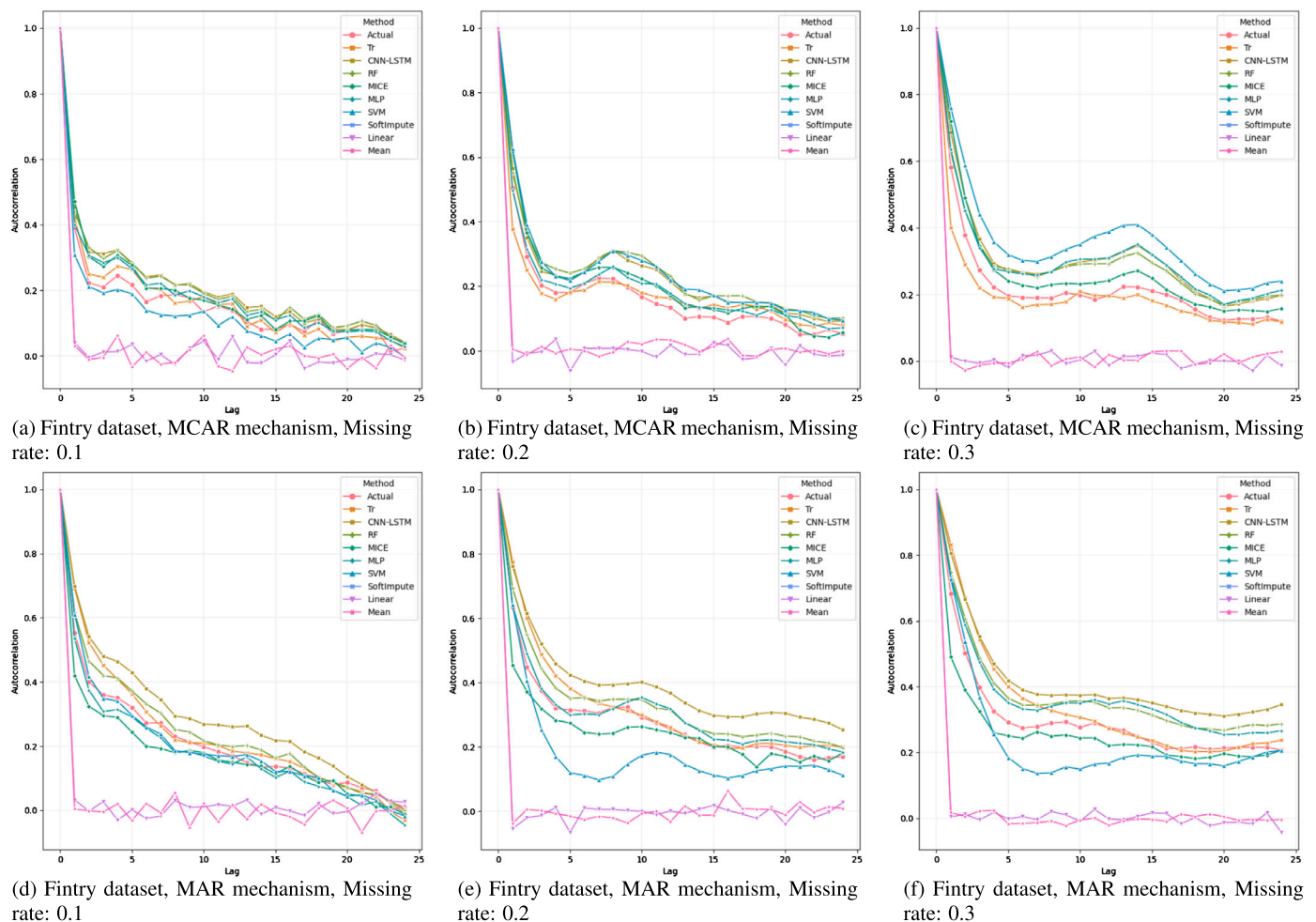


Fig. 13. Autocorrelation plot for two missingness mechanism and three different missing rates, Fintry dataset.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ashkan Lotfipoor reports financial support was provided by Heriot-Watt University.

Data availability

The authors do not have permission to share data.

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